

Regression methods

$$x \in \mathbb{R}^p, y \in \mathbb{R}$$

(low bias, high var., if...)

Method	Model paras.	Tuning paras.	High complexity if...	Fitting
k NN	—	$k = 1, \dots, m$	k small	— least squares
Linear reg.	$\vec{\beta} = (\beta_0, \dots, \beta_p) \in \mathbb{R}^{p+1}$	—	(p large)	LS with GD or closed sol.
Lasso/Ridge	$\vec{\beta} \in \mathbb{R}^{p+1}$	$\lambda \geq 0$	λ small	penalized LS with GD (lasso) or closed form (ridge)
Reg. tree	(splitting vars and splitting points)	tree depth $ T $ or cost-complexity para. $\alpha \geq 0$	$ T $ large/ α small	penalized LS
Bagging	— " —	—	—	LS
RF	— " —	m (nb. of features in each split)	(m large)	LS
Boosting	— " —	no. B, splits $d \in \mathbb{N}$ shrinkage $\alpha \in (0, 1)$	B large d large α small	LS

Neural networks

biases $\beta^L, \dots, \beta^{(M+1)}$
weights $W^L, \dots, W^{(M+1)}$

width, depth of NN
penalty params in fitting
filter sizes in CNN
etc.

width large
depth large
many FC layers
etc.

LS with
back-prop
and stochastic
 G

Classification Methods $x \in \mathbb{R}^p$, $y \in \mathcal{Y} = \{1, \dots, q\}$ classes

Method	Model paras	Tuning paras.	High complexity if...	Fitting
lin. NN	—	$h = 1, \dots, n$	h small	—
log. regr. (multinomial) + lasso/ridge	$\vec{\beta} \in \mathbb{R}^{p+1}$	— $\lambda \geq 0$	(p large) λ small	Maximum likelihood (ML) penalized ML
LDA	$\begin{cases} \pi_1, \dots, \pi_q \\ \mu_1, \dots, \mu_q \\ \Sigma \in \mathbb{R}^{p \times p} \end{cases}$	—	(p large)	ML
QDA	$\Sigma_1, \dots, \Sigma_q$	—	— " —	ML
Classification trees bagging / RF / boosting MMC / SVC / SVM	$\vec{\beta} \in \mathbb{R}^{p+1}$ for MMC $\rightarrow$ SVC $\alpha_1, \dots, \alpha_m$ weights	as in regression budget b SVM: tuning para of kernel (e.g. γ, τ, σ for poly. Gaussian kernel)	small b large γ	Gini index ML (variance / cross-entropy) coarse optimization

Neural Networks

same as for regression

minimizing the deviance
of multinomial (ML)

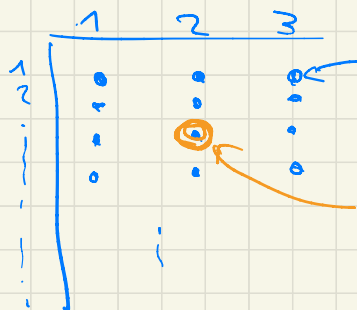
Cross-validation

• grid Search CV

e.g. in boosting

nb. of trees
 B

fits models for different tuning parameters on a "grid"



each point is a model fitted K times.
 $\Rightarrow$ compute CV error and
choose best one

best C error $d=2$, $B=3$

$\Rightarrow$ Fit model with $d=2$ and $B=3$
again to all training data.